

Electrical Fuse Schedule (Implementation - Lynx Topology)

As-of date: 2026-06-01

Purpose: define each required fuse by circuit, protected conductor/device, holder/housing method, physical placement, and linked wire-gauge assumptions for the approved Phase 1 Lynx architecture with a battery-backed 12V bus and dedicated 48V secondary alternator branch.

Related docs:

- Canonical electrical/system baseline: docs/core/SYSTEMS.md
- Implementation topology and conductor map: docs/implementation/ELECTRICAL_overview_diagram.md
- Decisions/open items tracker: docs/core/TRACKING.md
- BOM source of truth: bom/bom_estimated_items.csv

Design Basis

- Topology: Victron Lynx Distributor M10 (LYN060102010) with 3 active fused 48V branches and Lynx Slot 4 open/spare.
- Battery bank assumption: 3x 48V 100Ah batteries in parallel (3 separate battery-positive conductors leaving batteries).
- 12V distribution assumption: 12V fuse block used as the shared junction device (main + stud = source combine, integrated negative bus/main - = return), fed by Orion-Tr Smart 48/12-30 charger and a 12V 100Ah LiFeP04 buffer battery branch.
- Parallel-bank safety rule: use one Class T fuse per battery-positive conductor leaving the battery.
- Active branch devices on Lynx:
 1. MultiPlus-II 48/3000
 2. SmartSolar 150/45
 3. Dedicated 48V secondary alternator branch (Mechman/WS500 path)
 4. Slot 4 / F-05 open spare — Orion-Tr 48/12-30 input is **not** a Lynx fused branch; it uses standalone source-side F-06 from a Lynx 48V+ bus tap.
- Alternator path lock for this pass:
 1. F-04 is locked to 150A MEGA (58V/80V) in Lynx Slot 3 at the house-bank/Lynx end of the alternator positive run.
 2. Obsolete pre-Mechman charger/fuse paths are removed from active architecture and primary layout planning.
 3. WS500 required fused leads are F-12 regulator power and F-13 positive voltage sense; current-sense high/low is an unfused twisted sense pair per current Wakespeed manual.
 4. Ford Upfitter Switch #3 is locked as the manual WS500 enable source through F-15 .

Alternator Branch Fuse + Wire Finalization (2026-03-19)

Assumptions for this pass:

1. One-way alternator run length: 20 ft .
2. Dedicated positive and dedicated negative runs (loop length 40 ft).
3. Charging target voltage basis: 58.4V .
4. Voltage-drop planning target: about <=2% under expected charging current.

Resistance and drop screen ($V_{drop} = I * (2 * L * R_{per_ft})$):

Gauge	150A drop	% @ 58.4V	200A drop	% @ 58.4V	Decision note
4 AWG	1.49V	2.55%	1.99V	3.40%	Reject for this run length/current class
2 AWG	0.94V	1.61%	1.25V	2.14%	Acceptable for ~150A class
1/0 AWG	0.59V	1.01%	0.79V	1.35%	Strong margin
2/0 AWG	0.47V	0.80%	0.62V	1.07%	Best margin; inventory already on hand

Lock for this build pass:

- Keep the planned alternator branch fuse at 150A (F-04 , Lynx Slot 3).
- Reuse existing uncut 2/0 inventory for alternator + and dedicated negative run.
- Row 173 contingency purchase is removed from scope unless field measurement proves an actual shortfall.

Required Fuse Map (Start-to-Finish, With Housing)

Fuse ID	Circuit (source -> load)	Protected wire/device	Fuse type and voltage class	Amperage	Holder or housing method	Physical location	Physical notes
F-01A	Battery A + -> bank positive combine/disconnect input	Battery A positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment, within ~`7"`` of Battery A positive post	2/0 AWG
F-01B	Battery B + -> bank positive combine/disconnect input	Battery B positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment, within ~`7"`` of Battery B positive post	2/0 AWG
F-01C	Battery C + -> bank positive combine/disconnect input	Battery C positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment, within ~`7"`` of Battery C positive post	2/0 AWG
F-02	Lynx Slot 1 -> MultiPlus DC+	Main inverter positive feeder	MEGA, 58V or 80V	125A	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 1	2/0 AWG (AWG 1 on shunt)
F-03	Lynx Slot 2 -> SmartSolar BAT+	MPPT battery-side positive feeder	MEGA, 80V Victron replacement stock	60A	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 2	6 AWG
F-04	Lynx Slot 3 -> 48V alternator branch input (ALT+)	Alternator-to-house positive charge cable	MEGA, 58V or 80V	150A	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 3	2/0 AWG (lock)
F-05	Lynx Slot 4	Open spare fused position; no Orion connection in active topology	N/A	Not installed	Leave blank/open for now; legacy 40A MEGA stock remains spare-only and is not Orion protection	Lynx Distributor, Slot 4	N/A
F-06	Lynx 48V + bus tap -> inline fuse -> Orion 48V input +	Orion 48V input/device lead; this is the one source-side Orion input fuse	Interim/current: existing 30A 58V MIDI on the short 6 AWG Orion input run. Final cleanup stock purchased 2026-06-01: 3x Mouser 576-	Interim 30A; final 20A per Victron Orion 48V external battery-protection recommendation	Interim MIDI holder now acceptable for build/bench use; final FKS/ATO stock	Electrical cabinet between Lynx + bus tap and Orion input	6 AWG pigtail acceptable for the final holder direct 6 AWG

			166.7000.5202 / Littelfuse 166.7000.5202 FKS/ATO blade fuses (20A, 80VDC) plus 3x Mouser 576- 178.6150.0001 / Littelfuse 178.6150.0001 ATO/FKS holders (80VDC per owner- confirmed listing).		purchased in BOM row 182. Keep the unfused source-side tap from Lynx to F-06 physically short and protected.		
F-07	Orion 12V output + - > 12V fuse block main + stud	Main 12V feeder from Orion into shared source- combine point	MEGA, 80V Victron replacement stock	60A	Victron MEGA fuse holder (external, non-Lynx)	Electrical cabinet, within ~`7"of Orion12V` output stud	6 AWG AWG m per O table.
F-09A/B/C	PV string + leads -> MPPT PV combiner	Each solar string positive conductor and reverse-current path	gPV string fuse (>=150VDC)	15A each (provisional)	10x38 touch- safe PV fuse holders in weatherproof combiner enclosure	Roof-entry combiner near gland/pass- through	10 AWG
F-10	12V fuse block branch circuits -> each 12V load	Individual 12V branch conductors and load circuits	ATO/ATC blade fuses (32V class)	Per-circuit	Integrated sockets in marine 12V fuse block	12V fuse block in electrical cabinet	Per b
AUDIO-HU / 12V-12	12V fuse block -> Kicker 46KMC2 media center/source unit	KMC2 source/head- unit branch conductor and device harness	ATO/ATC branch fuse (32V class) plus KMC2 harness 15A ATM fuse	15A	Integrated 12V fuse block branch plus OEM KMC2 harness fuse	Electrical cabinet to driver-side DC shelf/source face	12 AWG kept ft; 10 route toward
AUDIO-SUB	12V source/main + stud -> external fuse -> Kicker 49PTRTP10 powered sub +	Powered-sub positive branch and subwoofer input; manual- required external protection	Inline/MRBF/AFS/ANL- class fuse matched to selected 4 AWG kit (32V+ DC class acceptable on 12V branch)	40A	Holder within about 18 in of 12V source takeoff; if using Kicker 47KMPK4, fit the PTRTP10- required 40A fuse rather than the kit generic larger fuse	Near 12V source/junction, downstream of SW-12V-BATT	4 AWG with n AWG re 12V r bus/n
F-11	12V buffer battery + -> 12V fuse block main + stud via SW- 12V-BATT	Buffer battery source cable and downstream junction fault exposure	Inline MIDI/AMI/ANL family rated >=32VDC	100A class baseline	Sealed inline holder mounted close to battery positive	Within ~`7" of 12V buffer battery positive post	4 AWG
F-12	WS500 regulator power lead	WS500 power feed lead	Inline fuse/holder rated for the actual source voltage;	10A baseline (15A if required by	Sealed inline holder per WS500	Near source end of WS500	Harne

			current Wakespeed manual recommends sealed ATC, but this build treats the lead as 48V-referenced unless final harness docs prove otherwise	extra-large alternator case)	kit/manual, with verified voltage rating	regulator power lead	
F-13	WS500 positive voltage-sense lead	WS500 battery/charging voltage sense lead	Inline fuse/holder rated for actual 48V bank maximum voltage unless WS500-supplied harness documentation proves otherwise	3A	Holder per WS500 manual/kit, voltage class verified before install	Near positive sense takeoff on charge side of the main alternator branch fuse / battery-bank connection	Harne
CERBO - PWR	48V system positive -> Cerbo GX power +	Cerbo GX low-current electronics feed	Inline fuse/holder rated for 48V bank maximum voltage	1A - 3A	Small inline holder close to the positive takeoff	Electrical cabinet, preferably system/load side of main disconnect so Cerbo powers down with the house system during bench shutdown	18 AWG duplex accept
F-15	Ford Upfitter #3 -> WS500 brown ignition/enable wire	Low-current regulator enable/control wire	Inline ATC/ATO (32V class acceptable; 12V control circuit)	3A	Sealed inline holder near the Ford upfitter blunt-cut source / splice handoff	Engine bay or control-wire handoff point before small-gauge run to WS500	16 AWG
OEM-SHUNT	Battery-side positive or Lynx/system positive tap -> SmartShunt Vbatt+ terminal	SmartShunt electronics supply/sense lead	External Victron-supplied red cable with inline low-current fuse; not an internal SmartShunt fuse	OEM value	Inline holder in supplied red cable	Prefer battery-side positive if SOC continuity is desired while the main disconnect is open; system side is acceptable if zero parasitic draw during disconnect-off storage matters more	OEM lead

Retired Fuse IDs

Obsolete pre-Mechman charger/fuse paths are removed from the active schedule. Do not reserve board space or labels for them.

Spare Fuse Inventory (Updated)

Fuse type	Installed qty	Spare qty to carry	Notes
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Class T 200A (provisional installed)	3	1 on hand	Owner confirmed 3 holders and 4 slow-blow Class T fuses total (3 installed + 1 spare); add more only if the one-spare-per-installed policy is later desired
MEGA 150A (58V/80V)	1	2	Alternator branch (F-04) installed + spare set (row 170)
MEGA 125A (58V/80V)	1	4	MultiPlus branch
MEGA 60A (80V)	2	3	Victron 5-pack row 188: install MPPT (F-03) + Orion output (F-07), keep 3 spares. Earlier 6x low-cost 60A MEGA batch was owner-confirmed misadvertised/not actually 58V and is quarantined/trash, not 48V install stock.
MEGA 40A (58V/80V)	0 active	3	Legacy spare stock only; do not use as Orion device protection
Orion 48v input protection	Interim installed/available: existing 30A 58v MIDI. Final cleanup stock purchased: 3x Mouser 576-166.7000.5202 20A 80VDC FKS/ATO fuses + 3x 576-178.6150.0001 80VDC holders	Install 1x final fuse/holder and carry 2 spare final 20A FKS/ATO fuses + holders	Interim 30A protects the short 6 AWG run; final 20A matches the Orion manual and is the preferred cleanup hardware
12V buffer battery main fuse (100A class)	1	3	Spare pack basis is BOM row 105
WS500 regulator power fuse (F-12)	1 active position	2	Carry 10A and 15A spares with holder/fuse voltage rating verified for the source voltage
WS500 positive voltage-sense fuse (F-13)	1 active position	2	Carry 3A spares with holder/fuse voltage rating verified for the 48v bank maximum
Cerbo GX power fuse (CERB0-PWR)	1 active position	1	Carry a spare 1A-3A fuse/holder rated for the 48v bank maximum
WS500 ignition/enable fuse (F-15)	1 active position	2	Carry spare 3A mini/ATO fuse and one spare sealed holder; 12V control circuit
PV string fuse 15A gPV	3	3	One spare per string
SmartShunt OEM harness fuse	1	1	Keep OEM-equivalent spare if field-replaceable
ATO/ATC branch fuses	variable	2 each used value	Keep mixed kit onboard
Camper audio KMC2 branch (AUDIO-HU)	1 active 15A branch plus KMC2 harness 15A ATM	2 spare 15A blade/ATM fuses	Preserve both source-side conductor protection and the KMC2 harness fuse
Camper audio powered sub (AUDIO-SUB)	1 active 40A external fuse	1-2 spare 40A fuses matching the selected holder family	PTRTP10 manual calls for 40A external fuse and 4 AWG power/ground

BOM Row Mapping

Fuse scope	BOM row(s)
Main battery Class T protection (F-01A/F-01B/F-01C) + Class T spares	bom/bom_estimated_items.csv row 7
Lynx branch MEGA fuses (F-02 to F-04 installed; Slot 4/F-05 open) + spare set	bom/bom_estimated_items.csv rows 10, 170, and 188
Orion installed fuse-holder hardware (F-06, F-07)	bom/bom_estimated_items.csv row 11
Orion input fuses (F-06 active; F-05 not used in active topology)	bom/bom_estimated_items.csv rows 11, 133, and 182
WS500 low-current fuse/holder kit (F-12, F-13)	bom/bom_estimated_items.csv row 171
Cerbo GX power feed (CERBO-PWR)	bom/bom_estimated_items.csv row 22; small inline fuse/holder may come from low-current install stock
WS500 Upfitter #3 enable/control path (F-15)	bom/bom_estimated_items.csv row 176
12V buffer battery (B12)	bom/bom_estimated_items.csv row 21
12V buffer battery main fuse + holder (F-11)	bom/bom_estimated_items.csv row 125
12V battery disconnect (SW-12V-BATT)	bom/bom_estimated_items.csv row 124
12V branch panel and blade fuses (F-10)	bom/bom_estimated_items.csv row 16
Camper audio source/head-unit branch (AUDIO-HU / 12V-12)	bom/bom_estimated_items.csv rows 189, 192, and 193
Camper audio powered-sub branch (AUDIO-SUB)	bom/bom_estimated_items.csv rows 191 and 192
SmartShunt external OEM fused red lead (OEM-SHUNT)	bom/bom_estimated_items.csv row 23 (included with SmartShunt kit)
PV string fuses + holder (F-09A/B/C) and spares	bom/bom_estimated_items.csv row 106

Assumptions and Open Items

- Wire sizing above assumes copper conductors and enclosed vehicle routing.
- F-09A/B/C and the old 3S3P fuse count are modeling placeholders only; final PV fusing waits until solar modules/stringing are selected after shore and alternator charging are working.
- Confirm Mechman kit content/polarity (PH / NH) and WS500 harness voltage ratings before final low-current fuse-holder purchase/install.
- SW-12V-BATT remains manual-only in Phase 1; no automatic LVD behavior is assumed.
- Final lock for F-11 still requires explicit 12V buffer battery/BMS continuous discharge-current confirmation.
- F-15 exists to protect the smaller-gauge control wire between Ford Upfitter #3 and the WS500 brown ignition/enable wire; the factory 25A upfitter circuit protection is not the final wire-protection device for that branch.
- Orion F-05/F-06 lock: the active topology uses F-06 as the single source-side Orion input fuse from a Lynx 48V+ bus tap. Lynx Slot 4/ F-05 stays open/blank because practical 20A Lynx/MEGA protection is not available; the existing 40A Lynx stock is spare-only and must not be installed as Orion device protection. Do not stack a Lynx MEGA fuse plus F-06 inline fuse for the Orion branch unless the topology is deliberately reopened.
- Camper audio 12V load: KMC2 source branch and PTRTP10 powered-sub branch are downstream 12V loads, not new 48V branches. The PTRTP10 40A branch can exceed Orion 30A continuous output during heavy bass; rely on the 12V buffer battery for transients and validate sustained loud-use behavior before assuming all 12V loads can run at max simultaneously.